Innovation Methods Down-selection Tools

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Learning objectives

Unit level learning objective(s)

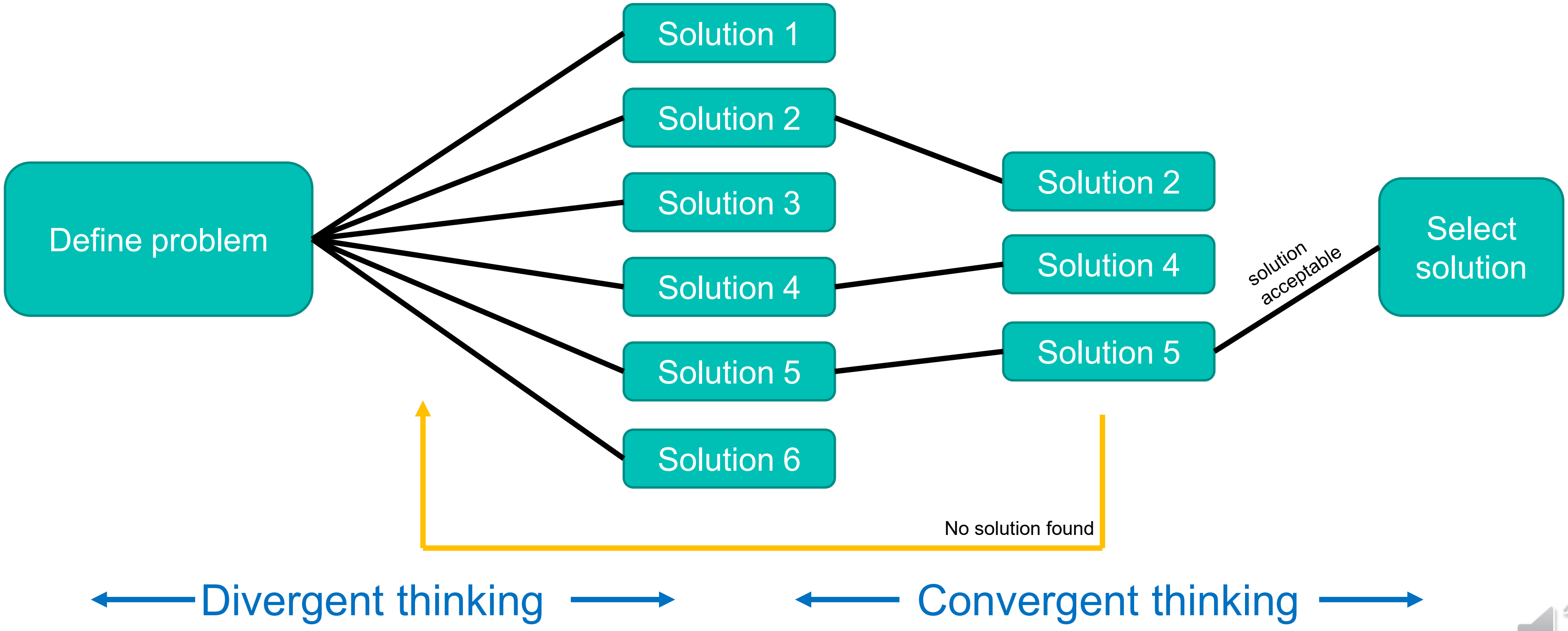
2. Carry out the design, build and test of a functioning major UAV assembly as part of a team, using applicable interdisciplinary concepts and methods;

Session level learning objectives

1. **Recall** the benefits of using a structured down-selection methodology
2. **Utilise** a *numerical* and *non-numerical pairwise comparison tool* to **rank** or **weight** solution criteria
3. **Utilise** a *controlled convergence* tool to **select** an optimum solution
4. **Utilise** a *multi-criteria decision analysis (MCDA) method* to **select** an optimum solution

↓ Higher Order

Divergent - convergent thinking



Reviewing requirements and solutions

Criteria / Requirements

Define, Select, Weight or Rank your criteria

i.e. weight, cost, time-to-market, capital investment, safety…

Solutions/Designs

Use your criteria to select your solution

i.e. design 1, design 2…

Tools

Evaluate criteria / requirements

Pairwise Comparison (ranking)

Pairwise Comparison (numerical)

Evaluate solutions/designs

Controlled convergence

*Multi-Criteria Decision Analysis** (MCDA)

*combines evaluation of criteria and solutions

Pairwise Comparison (ranking)

Pairwise comparison (ranking)

Function		A	B	C	D	E	F
Strength	A		AB	A	D	E	A
Appearance	B			C	E	E	F
Speed	C				C	C	F
Carbon footprint	D					D	F
Design resource	E						E
Durability	F						

	Count	Rank
Count of A	2.5	3rd
Count of B	0.5	5th
Count of C	3	=2nd
Count of D	2	4th
Count of E	4	1st
Count of F	3	=2nd

1. List your criteria
2. Compare 2 criteria and record the letter for the more important criteria. If equal, write both letters in the box
3. Repeat until all criteria are compared
4. Count the number of times the letter appears. Where 2 letters appear in a box only assign 0.5 to each letter.
5. Rank the criteria

Pairwise Comparison (numerical)

Pairwise comparison (numerical)

Feature or requirement for comparison	Details about that requirement
Initial Cost	Cost to purchase (not maintain/upgrade)
Durability	Lifespan (time), not distance
Weight	Mass when fully fuelled
Handling	Everyday driving
(empty)	
(empty)	
(empty)	

1. List your criteria
2. Assign a score to more important and much more important
3. Compare each criteria with the other
4. Extract the weighted (%) result and use in your discussion, or as input to other tool

Pairwise comparison matrix

	=Much More Important	***=More*** Important	The ***=Same***	***=Less*** Important	***=Much Less*** Important
Choose a scale (symmetric) →	9	3	1	0.333333	0.111111111

Criteria		**Initial Cost**	**Durability**	**Weight**	**Handling**	**(empty)**	**(empty)**	**(empty)**		
	X	1	2	3	4	5	6	7	Total	%
Initial Cost	1	X	3	9	0.333333				12.33	56
Durability	2	0.333333		3	3				6.33	29
Weight	3	0.111111			0.111111				0.22	1
Handling	4	3							3.00	14
(empty)	5					X			0.00	0
(empty)	6						X		0.00	0
(empty)	7							X	0.00	0
		3.44	3.00	12.00	3.44	0.00	0.00	0.00	21.89	100
		15.7	13.7	54.8	15.7	0.0	0.0	0.0	100	21.89

Tools

Evaluate criteria / requirements

Pairwise Comparison (ranking) ✔

Pairwise Comparison (numerical) ✔

Evaluate solutions/designs

Controlled convergence

*Multi-Criteria Decision Analysis** (MCDA)

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Controlled convergence

Controlled convergence

1. List your criteria
2. Choose a ‘Datum’ design. List the other designs
3. Compare each design against the datum.
   - If better than datum record +
   - If equal record =
   - If worse than datum record –
4. Sum the =, + and -
5. Rank the results

CONTROLLED CONVERGENCE PROFORMA

	DATUM Concept ↓					
	balloon	helicopter	anti-gravity shoes	seagulls	empty	empty
cost		-	-	+		
range		-	-	-		
safety		+	-	-		
altitude ceiling		-	-	-		
empty						
empty						
∑ +		1	0	1	0	0
∑ -		-3	-4	-3	0	0
∑ S		0	0	0	0	0
Net Score	(Zero)	-2	-4	-2	0	0
Rank (1=best)	1	2	3	2		

Multi-criteria Decision Analysis

MCDA

1. List your criteria
2. List the designs
3. Identify a scoring scale
4. Identify weightings for the criteria
5. Give a score to each design for each criteria
6. Highest sum product = best design

Criteria	Designs	Scoring scale (i.e. 1-5, 1-10)
Sustainability	Lithium ion cell	1
Reliability	Hydrogen PEM cell	2
Power	Flywheel	3
Safety	Thermal store	4
empty	Solid oxide fuel cell	5
empty	empty	
empty	empty	

MCDA matrix

Criteria	*Weightings*	*Designs →*	Lithium ion cell		Hydrogen PEM cell		Flywheel		Thermal store		Solid oxide fuel cell		empty		empty	
	↓ Must add to 100% ↓		score	weighted score	score	weighted score	score	weighted score	score	weighted score	score	weighted score	score	weighted score	score	weighted score
Sustainability	0.150	1	4	0.6	3	0.5	1	0.2	1	0.2	2	0.3		0.0		0.0
Reliability	0.300	2	1	0.3	2	0.6	3	0.9	2	0.6	0	0.0		0.0		0.0
Power	0.250	3	3	0.8	1	0.3	5	1.3	5	1.3	1	0.3		0.0		0.0
Safety	0.300	4	2	0.6	5	1.5	0	0.0	5	1.5	3	0.9		0.0		0.0
empty		5		0.0		0.0		0.0		0.0		0.0		0.0		0.0
empty		6		0.0		0.0		0.0		0.0		0.0		0.0		0.0
empty		7		0.0		0.0		0.0		0.0		0.0		0.0		0.0
	1															
		Total score for design →	2.3		2.8		2.3		3.5		1.5		0.0		0.0	

Tools

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Evaluate solutions/designs

Controlled convergence

*Multi-Criteria Decision Analysis** (MCDA)

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Using a structured approach

Benefits

Helps you find agreement faster by providing a systematic framework for discussion

Takes the "personalities out of the room"

Numbers can help decision making

Easy to explain the selection to stakeholders

Risks

Random argument and discussion can generate unexpected solutions

Maybe someone's passion for the topic should give extra weight to their view?

The tool is only as good as what you put into it   rubbish in = rubbish out

Avoid hiding behind the tool as a reason for your decision making

University of BRISTOL

Thanks for watching

Click the link below to take a short quiz on this topic.